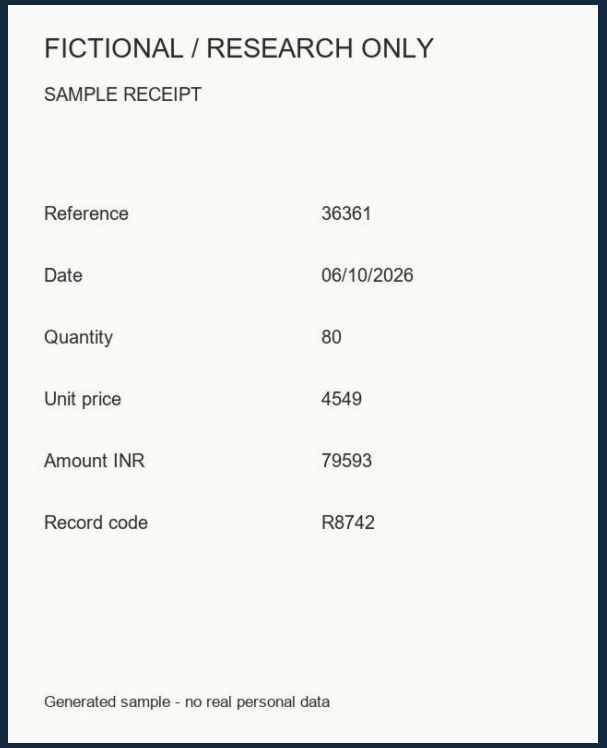
KIET Deemed to be University

DocGuard

Document tampering localization

CNN and Transformer comparison


FICTIONAL / RESEARCH ONLY

SAMPLE RECEIPT

Reference	36361
Date	06/10/2026
Quantity	80
Unit price	4549
Amount INR	79593
Record code	R8742

Generated sample - no real personal data

Synthetic receipt used in the experiment

Deep Learning and GenAI · CS312B
B.Tech CS · Semester 5 · Section D

The team

Deep Learning and GenAI · CS312B

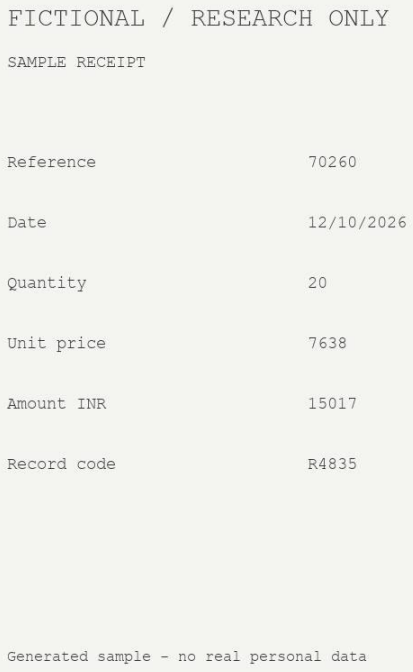
Shrashti Agarwal 2400290120240

Saumya Tiwari 2400240120230

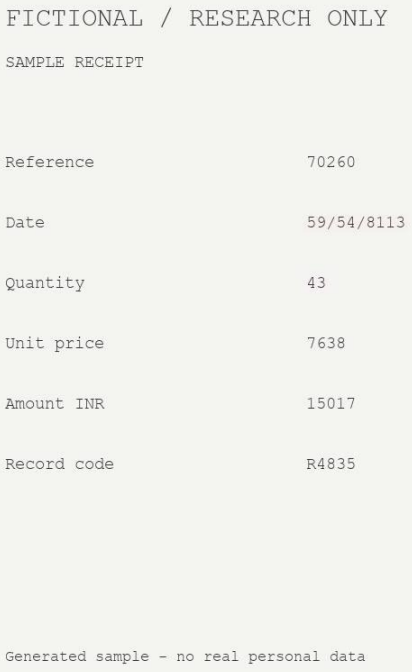
Subrat Singh 2400290120250

Where did the document change?

The review needs a location, not just a page label.



Original



Edited

Task

Predict a mask for suspicious regions.

Controlled study on fictional receipts.

The review pipeline

A local workflow from upload to downloadable evidence.

01 → **02** → **03** → **04**

Document

Image or PDF
Selected page

Preprocess

Normalize
160 × 200

Segment

CNN or MiT B0
Pixel scores

Review

Overlay and heatmap
PNG · PDF · ZIP

1,200 images. One controlled benchmark.

600 fictional source receipts, each with an authentic and edited version.

840
Training images

180
Validation images

180
Test images

Source identities and layout configurations are disjoint across splits.

Four fonts · six editable fields · rectangular edit masks
JPEG and blur acquisition variation applied to both labels

Two routes to a pixel mask

The same data split and output task; different feature extractors.

Compact CNN

30,081

parameters

Local feature extraction
Encoder and decoder with skip connections

MiT B0 Transformer

3,349,377

parameters

ImageNet pretrained encoder
Custom multiscale binary decoder

Training kept the comparison controlled

Preliminary CPU runs on a laptop.

3
Epochs per model

160 × 200
Input width × height

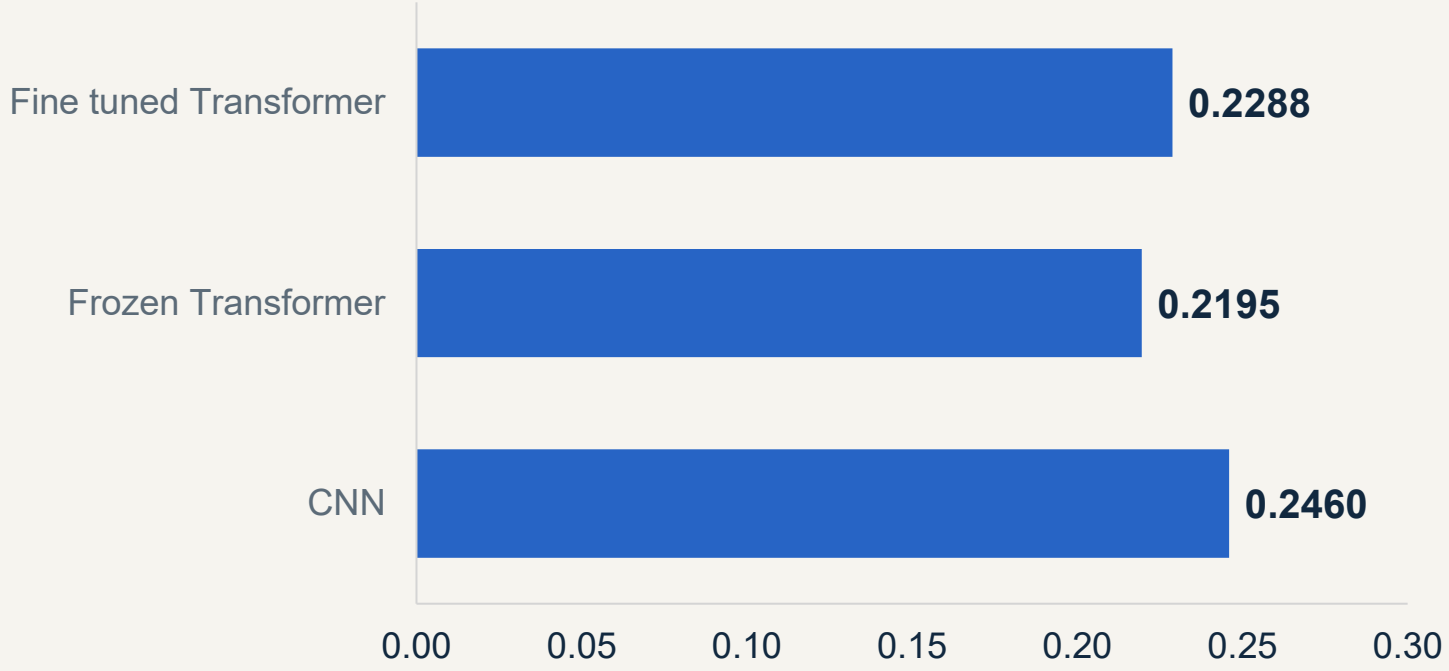
8
Batch size

Binary cross entropy + positive document Dice loss

Seed 42 · checkpoint and thresholds selected on validation data
Longer, higher resolution Colab experiments are planned, not executed.

CNN led the preliminary pixel F1 results

Higher is better. Global metrics on 180 held out test images.



0.2460

Best pixel F1
Compact CNN

Pretraining alone did not improve this experiment.

False positives remain the main barrier

Authentic pages incorrectly flagged by the current document rule.

90 / 90

CNN · 100% false flags

77 / 90

Fine tuned Transformer · 85.6%

Flag rule: predicted foreground ≥ 0.1% of the model grid.

Highlighted area is not a fraud probability.

The evidence behind the comparison

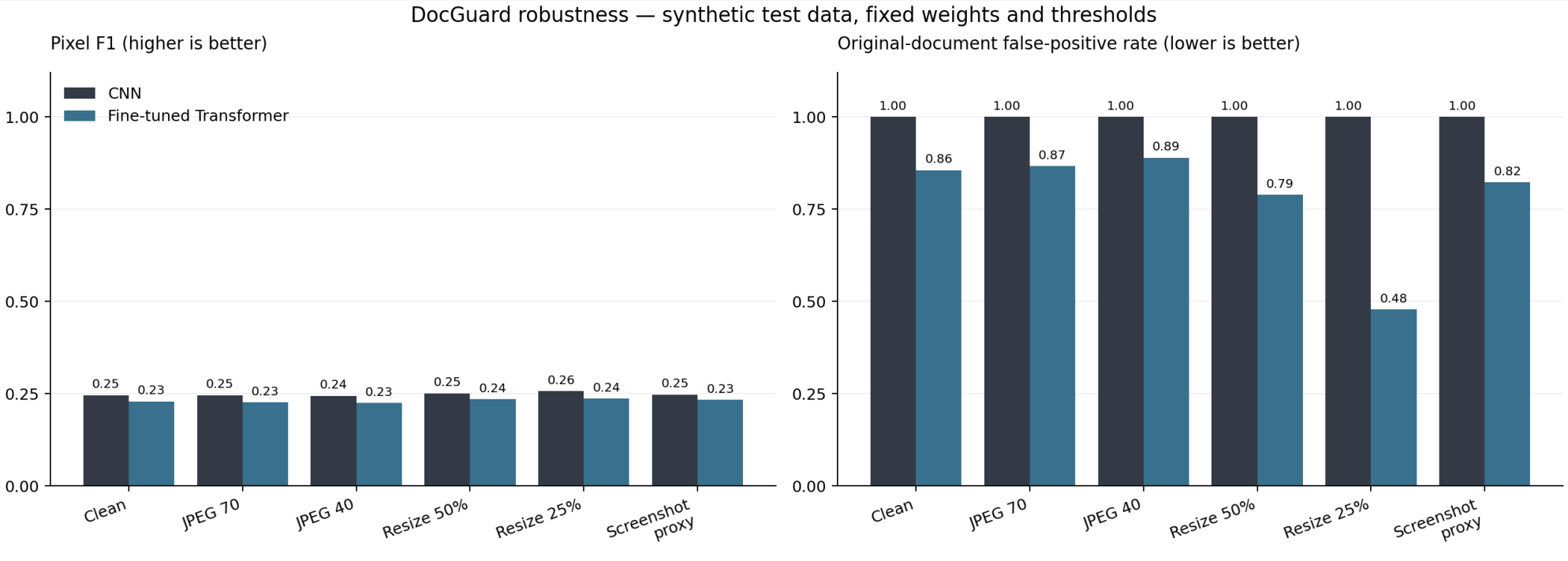
Precision, recall and IoU explain the weak localization results.

Model	Precision	Recall	F1	IoU
CNN	0.1878	0.3565	0.2460	0.1403
Frozen Transformer	0.1330	0.6283	0.2195	0.1233
Fine tuned Transformer	0.1691	0.3538	0.2288	0.1292

Low precision means many highlighted pixels are false positives.

Robustness across six conditions

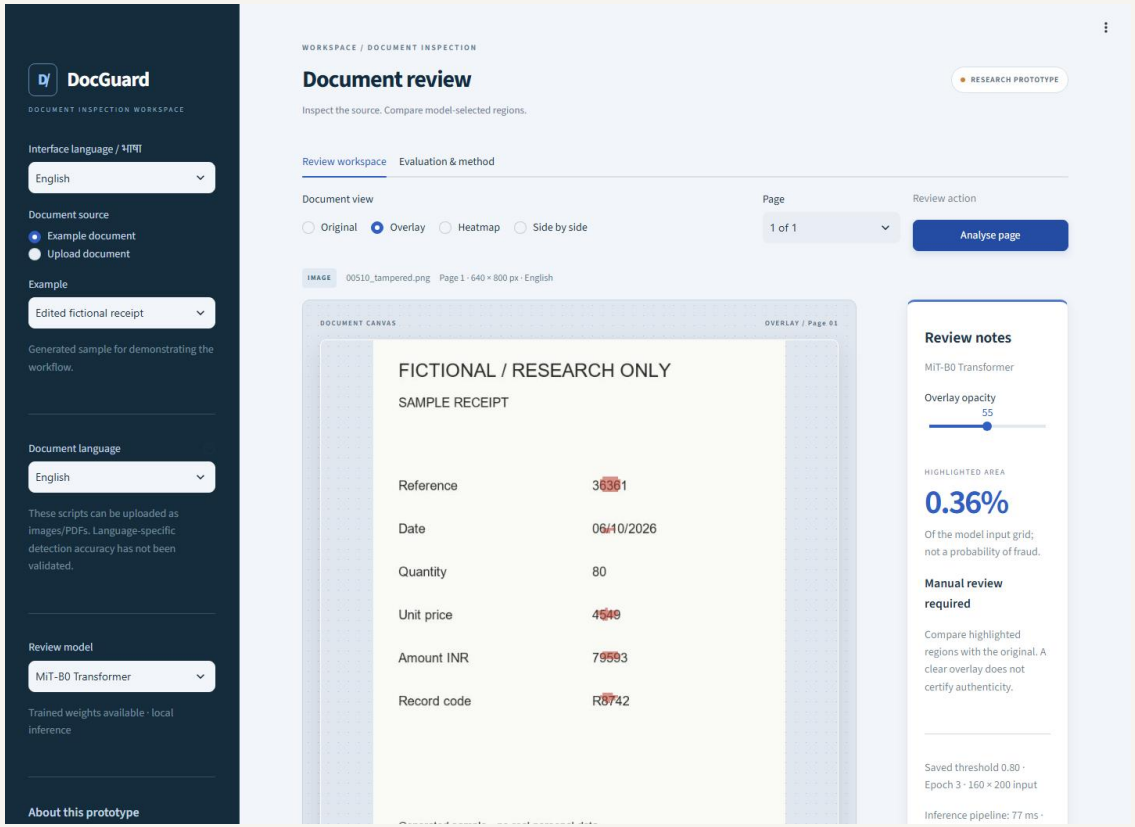
Fixed weights and validation thresholds; high false positives remain.



The screenshot condition is simulated. Small F1 gains do not establish reliability.

A complete document review workspace

Image and PDF upload · model comparison · exportable review evidence



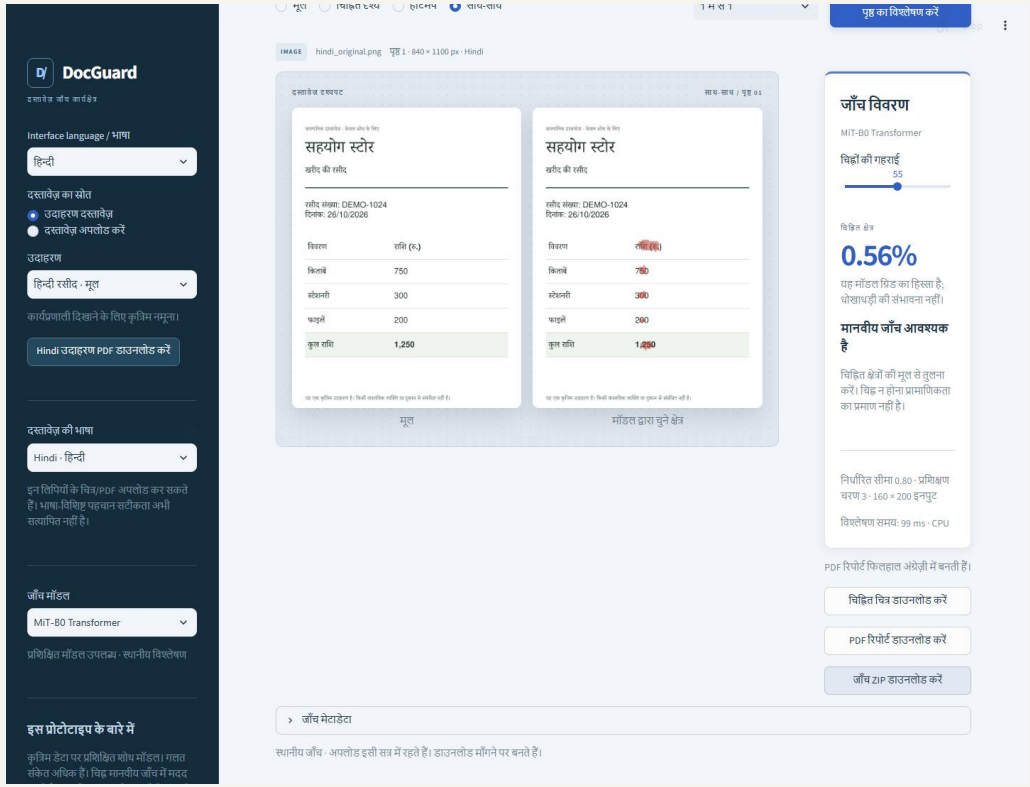
Local CPU

Overlay
Heatmap
Side by side

PNG
PDF
Review ZIP

Regional languages in one review

The current document and results stay available when the interface switches.



English
Hindi
Marathi
Tamil
Bengali

Hindi PDF examples included.
No OCR or translation.
Language accuracy is unvalidated.

Demonstration and next steps

A reproducible project with a measured, clearly stated limitation.

Live demonstration

Analyse an edited receipt
Compare both model overlays
Upload a Hindi PDF and export
Explain an authentic false positive

Next experiments

Licensed real documents
Higher resolution and longer training
Independent calibration
Real camera and scanner evaluation

14 unit checks passed · 14 browser flows · 7 language checks